Kailani Alarcon

Electrical Engineering · Robotics & Control Systems · Space Exploration

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RESEARCH INTERESTS

Space robotic manipulation and controls; autonomous and human-supervised robotic operations; sensor feedback and state estimation; simulation-to-test verification for lunar and planetary exploration tasks.

These are prospective research directions. See the proposed research questions; no externally reviewed publication is implied.

EDUCATION

University of Nevada, Reno — M.S., Electrical Engineering Expected May 2028

Focus: robotics, autonomous vehicles, and embedded systems

University of Nevada, Reno — B.S., Electrical Engineering 2026

Minor in Mathematics · Honors Baccalaureate

Relevant coursework: Robotics; Control Systems I & II; Digital Signal Processing; Embedded Systems; Microprocessor Applications.

ENGINEERING RESEARCH AND TECHNICAL EXPERIENCE

NASA Marshall Space Flight Center — OSTEM Intern Aug.–Dec. 2026

Human Factors Engineering & Mechanical Design · Huntsville, Alabama

- Independently carried out engineering assessment of a legacy robotic arm for a proposed high-vacuum lunar-dust test context, with NASA mentor oversight.
- Examined geometry, mass, actuator travel, workspace and operational constraints; used Python, Linux, Creo, ROS 2 and MuJoCo in evaluation and simulation work.
- Translated six operational scenarios into 34 hierarchical system requirements addressing manipulation, safety and verification.

Only previously public, résumé-level information is summarized here; confidential design materials and NASA approval are not implied.

LUX Dynamics — Product Development Intern Sep. 2023–Aug. 2026

Electrical, thermal and photometric validation · Reno, Nevada

- Performed validation work across 12+ LED products and developed thermal-testing procedures and Python/Arduino logging automation.
- Managed 4–5 concurrent development efforts and communicated technical results to engineering teams and stakeholders.
- Documented thermal and optical/lens-related work in two internal technical white papers.

University of Nevada, Reno — Engineering Peer Tutor Aug. 2024–May 2026

Provided individualized support to approximately 12–15 students each week on circuits, embedded C, microcontrollers and real-time control.

SELECTED TECHNICAL INVESTIGATIONS

UR3e Robot Kinematics and Jacobian Analysis — Spring 2026 · Team of three

Contributed to a group analysis of Denavit–Hartenberg kinematics and singularities; the team documented Jacobian rank 5 at the zero-angle pose and rank 6 at another evaluated configuration. Individual mathematical derivations were not separately attributed.

IVerify Capstone — Systems Integration Lead — Spring 2026 · Team of four

Led integration of sensing, analog electronics, firmware and indication subsystems for an IV-tubing monitoring concept. A two-LED bench path was demonstrated; clinical performance was not validated.

Thermal Controller & Logger — 2025–2026

Built and documented a Python-based temperature logging and reporting workflow. Recovered software supports eight thermocouple channels; published display illustrations and offline tests are synthetic.

Control & Measurement Laboratories — Spring 2025

Compared measured vibration and control behavior with models, including a reported vibration feature near 49.3 Hz and discrepancies motivating additional validation.

TECHNICAL METHODS AND TOOLS

Robotics and controls: Forward kinematics, Jacobians, MATLAB/Simulink, ROS 2, MuJoCo, PID control, test planning.

Programming and embedded: Python, C, C++, STM32, UART, SPI, I²C, PWM, VHDL, Git, Arduino.

Instrumentation and analysis: Oscilloscopes, thermocouples, VNA/S-parameters, FFT, thermal and photometric validation; Creo and Thermal Desktop.

TECHNICAL WRITING AND PUBLICATION STATUS

Internal professional reports (2): Thermal validation of commercial LED luminaires; optical/lens performance in LED luminaires. Written in an engineering workplace context and not publicly distributed here.

> **Scholarly record:** No peer-reviewed journal or conference publication is claimed as of October 2026. Public project documentation is available through the engineering case studies.

SERVICE, HONORS AND PROFESSIONAL MEMBERSHIPS

Service: University of Nevada, Reno Honors College Ambassador (tours, recruitment and orientation); Engineering Peer Tutor.

Honors: Presidential Scholarship; TC Energy Scholarship; engineering department and electrical engineering outstanding service awards.

Memberships: IEEE, Society of Women Engineers (SWE), and American Society of Mechanical Engineers (ASME). Memberships are not represented as officer positions.